CRISPDM DOCUMENTATION - Group 1

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PROJECT TITLE: ***BOX OFFICE ANALYSIS FOR NEW MOVIE STUDIO***

- CRISP DM(Cross-Industry Standard Process for Data Mining) refers to a popular methodology used in Data Mining, Data Analytics, and Data Science projects.
- It provides a detailed structure and iterative workflow for tackling Data Science projects.
- Some of the key stages involved in the CRISP DM framework include:
    - Business Understanding
    - Data Understanding
    - Data Preparation
    - Modeling
    - Evaluation
    - Deployment
- We will use the CRISP DM framework to analyze the Box Office content for a potential new movie studio.

Step 1: Business Understanding

Overview

With rising investments in original film material, the entertainment sector has grown immensely. Knowing the factors that contribute to success in the box office industry is paramount as our company prepares to open a new film studio. To identify trends about genres, release years, and ratings that support strong box office returns in revenue, this study examines historical movie data. The objective is to maximize audience engagement and profitability by directing strategic choices for film production and release.

Problem Statement

The organization seeks to determine which factors most influence film commercial success and derive insights into blockbuster film production, marketing (domestic and international), and release strategies.

Objectives

1. To analyze the box office database and datasets.
2. Determine the movie genres (e.g., Action, Adventure, Comedy) that perform the best.
3. To examine domestic vs international box office trends
4. To perform Exploratory Data Analysis (EDA) and identify key attributes that influence box office success
5. To provide actionable insights and recommendations for a robust film production strategy.

Challenges

Some of the key challenges faced in analyzing the box office industry and establishing a film studio include:

- Uncertainty in establishing the genres, budgets, and release timings that lead to box office success
- Lack of internal historical data and experience in film production
- The need to differentiate between domestic and international market dynamics.

Proposed Solution

This project aims to identify the factors most associated with box office breakthroughs by analyzing data from different sources such as Box Office Mojo, IMDb, and The Numbers. The solution entails developing a comprehensive pipeline—from data gathering, cleaning, and Exploratory Data Analysis (EDA) to extracting data-driven insights and recommendations that will support robust decision-making.

Conclusion

This analysis will empower the new studio with strategic recommendations on genre selection, ideal release timing, and target audience segmentation, maximizing the chances of producing profitable films and generating decent revenue, both domestically and globally.

Step 2: Data Understanding

- The dataset was collected from a variety of sources, which include:
  I. **Box Office Mojo**
     - This provided the gross earnings data, helping us understand the revenue generated by movies.
     - It provided detailed information about a movie's domestic and international box office revenues.

  II. **IMDb**
     - It offered information on genres, movie titles, release dates, runtime, and audience ratings.
     - This helped us in inspecting the relation of movie content and quality to box office success.

  III. **The Numbers**
     - This provided production budget estimates and deeper financial metrics/statistics
     - This was critical in establishing the total spending in movie production, and whether it paid off.

- The main datasets we have worked with in our analysis include:
  - **Bom.movie_gross.csv:** Provided the gross earnings (domestic and International), as well as studio details and release time
  - **im.db SQLite database:**
    - **Movie_basics table:** Contained movie titles, genres, runtime, and release year
    - **Movie_ratings table:** Contained average IMDb ratings and number of votes from audiences.

- Some of the key attributes used in our analysis include:
  - **Movie title:** the title of the movie
  - **Genre(s):** the specific movie genre
  - **Box office earnings:** gross earnings, both domestic and foreign
  - **Release year:** year of movie release

    - **IMDb rating:** average movie rating
    - **Number of votes:** number of votes per movie from the audience.

Data Checks

- Checked for missing values in the movies and financial data.
- Checked for duplicates in our datasets.
- Checked the data types of each column in the datasets.
- Checked for uniformity of data in the datasets.

Step 3: Data Preparation

- In our data preparation, we aimed to clean and transform the data to obtain deeper insights that would aid in providing actionable recommendations.
- Some of the key data preprocessing steps we undertook include:
    - **Selecting the data to merge**: we merged two tables from the SQLite database (movie_basics and movie_ratings) to have the movie titles and ratings in one table
    - **Handling missing values:** we filled some of the numerical columns with the median, and some of the categorical columns with the string 'unknown'. In addition, we dropped some rows from columns that registered a low number of missing values.
    - **Standardizing column names:** we ensured that all the columns were uniform across the datasets to ensure uniformity and readability. Some of the methods we used include:
        - strip() -> removes trailing whitespaces in a string
        - lower() -> converts a string to lowercase
        - replace() -> replaces characters in a string
    - **Feature Engineering:** in the movie_gross dataset, we added a column called **total_gross**, which resulted from the sum of **domestic_gross** and **foreign_gross**
    - **Plotting the distributions:** we performed EDA by plotting some of the key attributes to gain visual insights into the data we were working with. Some of the plots we did include:
        - **Average movie ratings vs Runtime minutes**: this was a scatter plot that aimed at establishing a correlation between the movie runtime and its average rating
        - **Top 20 movie genres:** this bar plot aimed at investigating the genres that had the highest number of movies. From the plot,

the Drama genre seemed to have the highest number of movies.

- **Top 20 Movies by Total Gross:** this bar plot aimed at inspecting the variation in total gross for different movies. From the plot, **Marvel's The Avengers** seemed to have the highest total gross, recording up to 1.5 billion USD in revenue.

Step 4: <u>Modeling</u>

This project did not entail the development of models for further analysis.

Step 5: <u>Evaluation</u>

- This stage entailed the formation of hypothesis tests to evaluate our findings and draw further insights.
- Some of the questions that led to the formation of hypothesis tests include:
    - Do movies from major studios (e.g., Warner Bros, Universal) consistently earn higher box office revenue than smaller studios?
    - Do some genres consistently earn higher ratings than others?
    - Do longer movies tend to receive higher ratings?

Insights

- The main insights that were obtained from these tests are:
    - Movies from major studios tend to generate higher revenue than movies from smaller studios
    - There is a significant disparity in ratings between genres
    - There is no significant correlation between movie runtime and average rating

Step 6: <u>Deployment</u>

- The analysis was carried out in a Jupyter Notebook.
- It was then saved and uploaded to the cloud (GitHub).